Online Resource 1

Quantity dispersion and reported price deviations in European carbon auctions

Conditional associations and historical scope

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S1 Purpose and fixed numerical record

This resource documents the separate quantity and price analyses and their limits. Sections S1–S11 retain the historical quantity evidence and subsequent sensitivity checks. Sections S12–S17 cover the matched German price analysis, including sources, descriptions and uncertainty. Section S18 provides the data-access route. All numerical estimates are preserved from the versioned analyses; the present reorganization changes no model or sample.

The repository is 124-creator/-123. The input is fixed at commit 5745eb3ddca79602346053446583d794b0e09800. The principal results are fixed at e25c1e02e1a58e0aec1084f7419066855e9565ec, and the robustness extension at d4b26f1bbd8c1d7c036259e48e119c9c12f6c9a9. The respective paths are:

- research/pro_review_20261010/full_data/
- research/eex_abc_results_20261010/
- research/eex_robustness_20261010/

The first path contains the raw-workbook identifiers, complete extraction, candidate sample, exclusions and cell traces. The other paths contain the analysis code, protocols, result summaries and computational records. Reproduction starts from the fixed input and rebuilds the eligibility rules rather than filtering until a particular sample size is reached. Source acquisition and redistribution remain subject to provider rights; the access route is described in Section S18.

S2 Fields, observations and eligibility

Table S1 Reported fields and research interpretation

Symbol	Original field label	Interpretation retained here
μ^b	Average volume bid per bidder	Reported bid mean
s^b	Standard deviation of bid volume per bidder	Reported bid SD; population and divisor unresolved
μ^w	Average volume won per bidder	Reported award mean
s^w	Standard deviation of volume won per bidder	Reported award SD; population and divisor unresolved
V	Auction Volume tCO2	Reported quantity, not a separately observed settlement total
B	Total Amount of Bids	Total reported bid volume
R	Cover Ratio	Reported demand coverage
N	Total Number of Bidders	Participants, not number of submitted orders
S	Number of Successful Bidders	Successful participants, not number of successful orders

Note: Quantity fields use tCO2 in the source. Literal labels are reproduced from row 6 of Primary Market Auction in each annual workbook. In particular, column M is “Total Amount of Bids” within the volume section; auction revenue is a different field. Original labels, cell positions and formats are retained in the provenance record. A reporting participant need not equal a final client or compliance installation. Field labels and arithmetic consistency do not establish identical populations for each paired mean and standard deviation.

The six annual files yield 1,327 dated observations. Restricting to T3PA, EU/DE/PL and explicit successful status yields 1,281 unique auctions. Exclusions comprise 38 other-product records, five other-series records and three cancelled auctions. Cancelled award fields remain missing rather than becoming zero dispersion. No main-sample row is removed on the basis of the size of its reported ratio.

The source audit compared 13 numeric fields for all dated observations with the extraction and the cell trace, with 17,251 comparisons in each route. An independent XML route agreed with the workbook import. The 53 candidate fields match their corresponding extraction rows; input files were unchanged after computation. These checks certify the relationship between the supplied files and the calculations, not unobserved bidder identities, every market event or historical first-release values.

S3 Projection and resampling implementation

The main nuisance space contains log cover ratio, log auction volume, log total bidders, log successful bidders, reporting-series indicators and original year–month indicators. Each auction has equal weight. Numerically stable least-squares projection gives a rank-79 pooled design with 1,202 residual degrees of freedom. The within-month implementation was cross-checked using an explicit dummy design and explicit row replication on fixed draws. Such agreement verifies a numerical route, not a second empirical sample.

For pooled block length L, draw starting months uniformly from $0, \ldots, M - L$ with $M = 72$. Append each selected contiguous block of L calendar months until at least 72 positions are obtained, then truncate. Retain all auction-series observations in each selected month. Multiplicities act as observation weights; original year–month labels are not renamed. Empty factor levels in a draw are removed rather than represented by all-zero dummy columns. Nuisance coefficients and the focal coefficient are re-estimated in each draw, and failures are recorded without replacement draws.

The principal stream uses SeedSequence([20261010, block_length]). The three-month analysis uses 999 draws shared across A/B/C; the one- and six-month checks use 199 each for A and C. Quantiles are computed by linear interpolation. The prior team calculation used a different initialization; this resource retains the frozen principal intervals rather than the more favourable of alternative Monte Carlo realizations. In particular, the principal ordinary-log slope interval includes one.

For the historical split, each original half has 36 months. Independent three-month resampling within each half produces 999 paired comparisons, shared across A and C. The extension protocol records the period stream [20261010, 2, 1], the reduced-control stream [20261010, 2, 2] and the series stream [20261010, 2, 3]. All nuisance slopes are separately estimated in each half. The two 97.5% slope-difference intervals constitute a Bonferroni-style sensitivity calculation only to the extent that each interval approximation is adequate. They do not correct selection of the historical question or fully represent cross-boundary dependence.

Reduced-control specifications use 499 draws, and within-series analyses use 399. Each subseries retains the full 72-month calendar axis, including months without selected auctions. EU and PL each have one such empty month. No observation is fabricated there. Singleton months contribute no net within-month information. The completed runs report no failed bootstrap draw. Bootstrap coverage remains approximate, particularly under changes in slopes or dependence.

S4 Additional numerical results

Table S2 Pooled block-length sensitivity

Spec.	Block months	Draws	β: 95% interval	θ: 95% interval
A	1	199	[0.7807, 0.9833]	[0.5903, 0.7158]
A	3	999	[0.7506, 1.0384]	[0.5714, 0.7422]
A	6	199	[0.7350, 1.0292]	[0.5564, 0.7438]
C	1	199	[0.7907, 0.9714]	[0.5855, 0.7134]
C	3	999	[0.7627, 1.0143]	[0.5660, 0.7402]
C	6	199	[0.7503, 1.0090]	[0.5526, 0.7493]

Note: These are exploratory percentile intervals. Three-month rows reproduce the principal result; they are not additional replications. Sensitivity endpoints based on 199 draws have non-negligible Monte Carlo variability.

Table S3 Influence sensitivity of residual correlation

Deletion diagnostic	A: θ range / value	C: θ range / value
One calendar month	[0.6419, 0.6656]	[0.6405, 0.6674]
One calendar year	[0.6058, 0.6981]	[0.6023, 0.6996]
One reporting series	[0.6019, 0.7120]	[0.6110, 0.7081]
One observation	[0.6467, 0.6569]	[0.6457, 0.6577]
Top 13 Cook-distance rows	0.6911	0.6930

Note: Bracketed values are deterministic deletion ranges, not confidence intervals. The largest-Cook-distance set is selected within each specification only for diagnosis. It does not define a revised data-cleaning rule. All main estimates retain 1,281 auctions.

Table S4 Within-series residual association

Spec.	Series	n	Residual df	θ [95% interval]
A	EU	840	764	0.6898 [0.6283, 0.7585]
A	DE	275	198	0.5619 [0.4567, 0.6808]
A	PL	166	90	0.8076 [0.6943, 0.8828]
C	EU	840	764	0.6839 [0.6159, 0.7587]
C	DE	275	198	0.5689 [0.4679, 0.6870]
C	PL	166	90	0.8138 [0.7005, 0.8850]

Note: Each series has its own month effects and numeric controls. Intervals use 399 three-month block draws. These are not independent-market replications or formal tests that series coefficients differ. The small PL residual sample limits interpretation.

The interquartile-scale contrasts reported in the main article are already present in the frozen extension output. Under A, $IQR\left(M_Z x\right)=0.0927037454$ and the corresponding fitted transformed-response contrast is 0.0816007553. Under C, these values are 0.1600483679 and 0.1407987656. They are partial-regression scale summaries. Exponentiating a transformed fitted value would not, without further assumptions, recover the arithmetic conditional mean of the original ratio. No such raw-scale mean or procurement outcome is estimated here.

S5 Denominator identities and rounding

For this diagnostic only, a superscript *c* labels constructed fields. Define $\mu^{b,c}=B/N$, $\mu^{w,c}=V/S$ and $R^c=B/V$, retaining V as the reported auction-volume field. For positive components,

$$\log r^{b,c}=\log s^b+\log N-\log R^c-\log V, \tag{S1}$$

$$\log r^{w,c}=\log s^w+\log S-\log V. \tag{S2}$$

If the common nuisance space Z^c contains these constructed log totals and counts, linear projection gives

$$M_{Z^c}\log r^{b,c}=M_{Z^c}\log s^b, M_{Z^c}\log r^{w,c}=M_{Z^c}\log s^w. \tag{S3}$$

The completed calculation yields maximum residual discrepancies of approximately 7.47×10^{-15} and 7.09×10^{-15}. These are algebraic and implementation checks, not another economic finding. They do not supply an independently observed total award W or authenticate $V = W$.

With the original reported rather than constructed values, mean discrepancies relative to B/N and V/S reach 0.5 quantity units, while the cover-ratio discrepancy relative to B/V reaches approximately 0.004994. Agreement with display precision is not certification of the historical rounding rule. The main regressions use the reported fields; neither implied means nor implied cover ratios silently replace them. The corresponding log-one-plus terms do not satisfy the additive log identities.

A related interpretation boundary concerns concentration. If m quantities genuinely form one population with common mean μ and $s^2 = \sum_i (q_i - \mu)^2/(m - d)$, then its single-auction quantity-share concentration is

$$H = \frac{1}{m} + \frac{m - d}{m^2}\left(\frac{s}{\mu}\right)^2. \tag{S4}$$

This conditional moment identity is standard algebra. It does not justify substituting N or S when the standard-deviation population is unknown. Cross-auction concentration additionally requires identity links across auctions. The current study neither computes a certified bidder-population concentration index nor infers matched-population compression from its cross-auction regression.

S6 Exploration record and author responsibilities

The snapshot, target and some results were inspected before the supplementary protocols were fixed. The first-freeze record is partial. The manuscripts and intervals cannot therefore be described as preregistered, selection-adjusted or based on an untouched test sample. The principal and robustness outputs are frozen for the current revision; no result was recomputed to make the text more favourable.

An earlier supplementary protocol proposed 0.10 as an exploratory reference for the absolute standardized association. It was introduced after earlier results were known and is not an economic-cost or journal threshold. It is not used as a decision boundary in the article's figures or as proof of publication value. This presentation change leaves the numerical record unchanged.

The numerical reimplementations use the same archives and are not independent-market replications. Fitting counts record computation rather than observations. The study received no financial support. Zhongfei Tian coordinated manuscript preparation and the final revision. Xiang Wang reviewed and checked the manuscript, approved its scientific content, and authorized completion of the final editorial revision. All inferential limitations remain part of the published analysis.

S7 Interpretation of the fitted quantities

The quantity coefficients, intervals and contrasts come from the frozen quantity outputs. Residual correlation is a symmetric in-sample statistic; the chosen direction of the regression is not evidence of temporal transmission. With one added regressor and identical residualization, partial R^2 equals the square of the residual correlation. These two numbers are not independent confirmations.

The reported fitted contrasts across the interquartile range of residualized bid dispersion remain on their respective transformed-response scales. No back-transformed conditional arithmetic mean,

representative bidder outcome or monetary benefit is inferred from them. Subperiod slopes and correlations use separately estimated nuisance coefficients and residual variances. Their difference is a descriptive historical comparison, not a structural invariance test robust to arbitrary changes in reporting or covariate distributions.

S8 Historical descriptions and direct correlation contrasts

Table S5 Descriptive statistics by historical period

Panel A: 2020–2022, n = 643

Variable	Mean	Q1	Median	Q3	Min.	Max.
Bid ratio r^b	1.532	1.345	1.520	1.712	0.874	2.622
Award ratio r^w	1.459	1.247	1.443	1.629	0.443	2.707
Cover ratio R	1.864	1.530	1.790	2.100	1.010	3.770
Auction volume V	2.692	2.363	2.515	3.082	0.972	6.399
Submitted volume B	4.844	3.887	4.761	5.588	1.937	12.899
Total bidders N	22.379	20	22	25	12	31
Successful bidders S	16.687	14	17	20	3	27

Panel B: 2023–2025, n = 638

Variable	Mean	Q1	Median	Q3	Min.	Max.
Bid ratio r^b	1.275	1.108	1.255	1.429	0.847	1.896
Award ratio r^w	1.384	1.184	1.356	1.557	0.665	2.295
Cover ratio R	1.785	1.550	1.710	1.980	1.080	3.780
Auction volume V	2.659	2.147	3.035	3.245	0.970	3.350
Submitted volume B	4.588	4.076	4.712	5.156	1.946	6.731
Total bidders N	22.948	21	23	25	15	30
Successful bidders S	17.071	15	17	20	5	27

Note: These previously tabulated summaries use the unchanged eligible auctions and equal row weights. Volume fields are in million tCO2; bidder counts and ratios are on their original scales. Quartiles use linear interpolation at $(n-1)q$ (type 7). No *p*-values, distributional-equality tests, trimming or new regression are introduced. These are marginal auction-level summaries; the early and late auctions do not represent matched bidder populations. Lower marginal ratios cannot be equated with a decline in identified bidder-level inequality, and the table does not explain the change in conditional association.

The direct differences in correlation in main Table 6, Panel C, already exist in the frozen extension: period_intervals.A.delta_theta_CI and period_intervals.C.delta_theta_CI in support/frozen_extension_results.json, and equivalently periods.A.delta_theta_CI95 and periods.C.delta_theta_CI95 in the repository's KEY_RESULTS.json. The point differences are 0.2296942521 and 0.2305336591. These are secondary exploratory 95% intervals, not the 97.5%

primary slope intervals. Displaying them now does not retrospectively change the extension protocol.

The numerical source ledger and aggregate audit are provided with the versioned manuscript package. They link displayed price values to fixed results and stored test statistics. The underlying source-input checks and estimation code are documented separately; an editorial ledger does not replace access to event-level inputs.

S9 Conditioning-set values and supplementary Figure S1

Table S6 Residual association across conditioning sets

Conditioning set	**A: $\hat{\theta}$ [95% interval]**	**C: $\hat{\theta}$ [95% interval]**
Series and year–month effects	0.6243 [0.5596, 0.7067]	0.6197 [0.5494, 0.7017]
Effects + volume + total bidders	0.6289 [0.5626, 0.7094]	0.6239 [0.5521, 0.7044]
Full controls	0.6496 [0.5714, 0.7422]	0.6486 [0.5660, 0.7402]

Note: Reduced-control intervals use 499 three-month block draws. The full-control row reuses the 999-draw principal intervals. Each conditioning set defines a different descriptive projection; these are not instruments or randomized comparisons.

Table S6 and Figure S1 display the same estimates; their agreement is not an additional result. The full-control row reproduces the principal quantity estimate. Main Figure 1 shows quartile ranges, main Figure 2 shows price-model CR1-t intervals, and main Figure 3 shows the secondary historical correlation contrast. These statistical objects are not interchangeable.

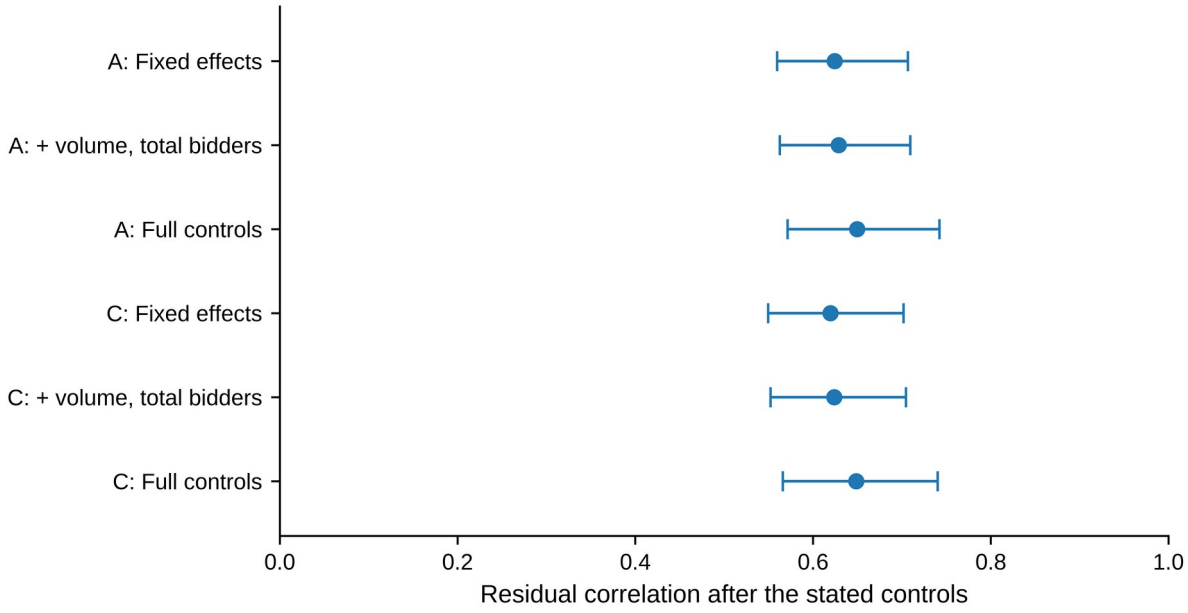


Fig. S1 Residual quantity correlations across conditioning sets; row labels identify the transformations. Bars are exploratory 95% three-month block intervals from Table S6. FE denotes reporting-series and year–month fixed effects; full controls add demand coverage and successful bidders to volume and total bidders. The full-control rows repeat the principal estimates

S10 Composition-standardization protocol and results

The supplementary protocol was recorded at commit 2151017871416fe0ad2505cfdbda23162aaa873e before the weighted fits, after the historical data

and preceding results had been examined. It is not retrospective preregistration. The E1 implementation and outputs are frozen at 89d3ac3beba670838d4b62b1209f6f3412753f19. The original equal-weight estimates and intervals are retained; newly generated unweighted intervals serve paired comparisons within E1 only.

The period model has 22 columns: an intercept, four standardized log controls, four squares, two nonreference-series indicators and eleven month-of-year indicators. It excludes year and year–month identifiers. Exact selected-term mean balance is a property of the fitted unpenalized logistic overlap weights, not a new empirical finding (Li et al. 2018). It does not authenticate full distributional balance, assignment ignorability or minimum variance for a residual-correlation contrast. The outcome model is re-estimated separately in each half with original year–month effects. Early weights equal e and late weights equal 1−e; resampling multiplies these by row multiplicity.

Table S7 Weighted contrasts and paired changes

Spec.	Contrast	Estimate	95% interval	97.5% interval
A	Weighted difference	0.1971	[0.1084, 0.2910]	[0.0947, 0.3029]
A	Weighting-induced change	-0.0326	[-0.1349, 0.0802]	[-0.1520, 0.0915]
C	Weighted difference	0.1937	[0.0982, 0.2843]	[0.0847, 0.3051]
C	Weighting-induced change	-0.0368	[-0.1449, 0.0797]	[-0.1668, 0.0942]

Note: Change means the overlap-weighted minus equal-weight late-minus-early correlation difference from the same bootstrap draw, not a mediation effect. The 0.10 interpretation reference in the supplementary protocol was introduced after earlier results were known; it is neither a cost threshold nor a journal criterion. The weighted-contrast 97.5% intervals include that reference.

Table S8 Block-length sensitivity of the weighted correlation difference

Block months	Tuples sampled	A: 97.5% interval	C: 97.5% interval
1	199	[0.0991, 0.3018]	[0.0916, 0.3033]
3	999	[0.0947, 0.3029]	[0.0847, 0.3051]
6	199	[0.1049, 0.2881]	[0.0996, 0.2780]

Note: Both period and outcome models are refitted in every draw. A and C share indices. No failed draw was replaced. Short Monte Carlo runs affect endpoint precision; absence of computational failures does not establish coverage. Weight ESS values of 586.55 and 614.75 are concentration diagnostics, not independent time-sample sizes. Maximum normalized weights are approximately 0.263% and 0.314%. The failed auction-volume distribution diagnostic is shown in main Table 7, not suppressed.

S11 Additional 2026 sample and finite block sensitivity

The E3 record reports a download on 10 October 2026 at 13:33:15 UTC; E3R retrieved the same URL at 14:12:19 UTC. Both records identify 72,742 bytes and SHA-256 54e6fb649872d229333adb29b62a3ff5b7fddd7393facc047050fb8f87a1bf3d. E3R rebuilt the January–September eligibility set, excluding seven October rows from 173 dated records and retaining 166 auctions; the point estimates matched E3. This is a repeated implementation on the same snapshot, not an independent market replication, first-release audit or certification of standard-deviation populations.

Table S9 The 2026 association and short-axis diagnostic ranges

Quantity / diagnostic	A	C
Slope β	1.3137	1.2375
Residual correlation θ	0.8494	0.8350
Original 999-draw θ interval	[0.8123, 0.9149]	[0.7832, 0.9156]
Non-circular finite θ percentiles	[0.8094, 0.9149]	[0.7757, 0.9156]
Circular finite θ percentiles	[0.8064, 0.9048]	[0.7731, 0.9055]
Leave-one-month-out θ range	[0.8396, 0.8626]	[0.8230, 0.8569]

Note: Point estimates and the original 999-draw intervals belong to E3. Finite-enumeration and deletion diagnostics are post-result E3R analyses. The enumeration percentiles are 2.5% and 97.5% quantiles of equally likely ordered block triples with linear interpolation. Deletion ranges are not confidence intervals. All schemes use the same 166 auctions and fixed models; they do not provide independent market replications.

For non-circular triples, the expected original-month multiplicities are (3,6,9,9,9,9,9,6,3)/7. Endpoint underrepresentation is therefore structural, not eliminated by 999 Monte Carlo draws. The 343 ordered triples yield 84 distinct month-weight vectors. The 729 circular triples yield 163 vectors and unit expected multiplicity for every month; they also connect September to January artificially. Both full finite distributions are retained. No claim is made that the circular scheme is universally more appropriate or that exact enumeration gives exact frequentist coverage.

E3R records 2,164 primary projection requests and eight explicit-dummy numerical cross-checks, with no failures and maximum implementation difference 4.60×10^{-14}. Its six synthetic checks assess enumeration, multiplicities and projection code. These execution counts are not observations. Complete outputs are at research/eex_integration_checks_20261010/results, commit 35d8c63fc57173f67eb0bd8bb570eb289188324f; they have not been re-estimated in this manuscript.

S12 German reported-price sources and reproduction boundary

The completed price study uses the German official-report comparison layer. Source-based matching identified 275 successful auctions in 24 quarters of 2020–2025, with price and quantity links to the EEX archive. The fixed restrictions quarantine 35 observations: 11 from 2022Q4, 11 from 2023Q4 and 12 from 2024Q2 for unresolved reference-footnote mapping, and one from 2025Q1 for a source-date conflict. This leaves 240 observations in 21 quarters, comprising 236 last-best-bid and four last-trade references. The predefined subset without row-level source flags contains 203 observations in 18 quarters. These qualification rules preceded the first price fit.

Price results and code are fixed at fccd471de96548bd781c59aced4a59435aae08ac under research/eua_local_price_execution_20261011. Descriptions and the targeted source-definition check are fixed at 33707c164c80accd7acf9fcd3f0965088a8370f6 under research/eua_price_description_followup_20261011. The analysis retains signed numeric percentage points, converted to basis points; it does not take the modulus of the source's EUR column merely because it is labelled Absolute. The local source check did not locate an explicit subtraction-and-denominator equation in the examined methods text. No rounded values are inverted to manufacture that definition. The scope is official-reported deviations, with native quote age and UTC event times unauthenticated.

The official PDFs and event-level price tables remain author-held; public aggregate outputs and code permit review but not an independent row-level refit. A hash alone is not an archive or a data-use permission. This is separate from the 2026 quantity snapshot, whose dynamic EEX URL and required hash are recorded in Section S11 but whose exact bytes are not included in the package. The additional 22 German 2026 price observations were acquired but not fitted, and are distinct from the 166-observation 2026 quantity exercise.

S13 Estimands and evidence status

Table S10 records the role of each analysis in the research sequence. Different transformations, uncertainty levels and later checks are not interchangeable confirmations. The labels describe recorded analysis roles, not a prospectively registered design.

Table S10 Evidence roles and limits

Analysis	Estimand	Uncertainty display	Interpretation
Historical pooled relation	Slope and residual correlation	Exploratory 95% month-block intervals	Within-sample conditional association
Historical period contrast	Primary slope difference; secondary correlation difference	97.5% slope intervals; 95% correlation intervals	Separate estimands; neither proves policy causality
Composition sensitivity	Weighted correlation difference	Exploratory 97.5% intervals; 95% paired-change intervals	Selected-moment comparison; incomplete volume-distribution balance
Additional 2026 period	Re-estimated slope and correlation	Nine-month diagnostics; finite percentile and deletion ranges	New-time description, not a frozen-model forecast
German reported-price relation	Signed-deviation coefficient and fitted interquartile contrast	CR1-t and CV3J intervals; one restricted wild-cluster test	Completed matched-subsample conditional analysis, not causal or predictive validation

Note: The price study has been completed and is documented below. Its one primary signed-outcome test and three secondary specifications are not confirmatory replications. No interval family adjusts the entire historical research process. Exact short-axis enumeration in the separate quantity extension does not imply exact repeated-sample coverage.

S14 Price estimand, conditioning set and descriptive context

The response is the official signed numeric percentage-point field multiplied by 100. Printed zero is retained at source precision; it is not a no-trade indicator or certification of identical unrounded prices. The focal covariate is log(1 + reported bid SD / reported bid mean). The fixed baseline contains an intercept, log auction volume, log total and successful bidder counts, log reported cover ratio, the submitted-price maximum minus minimum in EUR, log same-auction clearing price, observed-year indicators, and a last-trade indicator when it varies. The extended model adds only the focal covariate on the same rows. The primary matrix has rank 14 with 226 residual degrees of freedom. The 236-row best-bid model omits its constant reference indicator and has rank 13. The one-sided nature of a best bid, and the fact that reference prices are not midpoints, restrict the interpretation.

The source file for prices contains total bid volume B, not the reported cover-ratio field R. Eight shared quantities are cross-checked against the original annual EEX files. R is obtained directly from those original workbooks rather than replaced by B/V. No raw SD of awards is added to the price model, and no bid-to-award-to-price mediation relation is estimated. These are contemporaneous reported fields; the conditional comparison is not a pre-auction forecast.

Table S11 Reported price-deviation descriptions in the three fixed samples

Panel A: Signed deviations

Sample	*n* / *G*	Mean	Median	Q1–Q3	Range	SD
Primary	240 / 21	−2.80	−7.00	[−29.00, 17.25]	[−174.00, 171.00]	46.34
No row flags	203 / 18	−2.79	−5.00	[−27.00, 15.50]	[−174.00, 153.00]	43.23
Best bid only	236 / 21	−2.14	−7.00	[−29.00, 18.00]	[−174.00, 171.00]	46.19

Panel B: Absolute deviations

Sample	*n* / *G*	Mean	Median	Q1–Q3	Range	SD
Primary	240 / 21	33.51	25.00	[11.00, 46.00]	[0.00, 174.00]	32.06
No row flags	203 / 18	30.88	23.00	[10.00, 40.00]	[0.00, 174.00]	30.30
Best bid only	236 / 21	33.37	25.00	[11.00, 46.00]	[0.00, 174.00]	31.93

Note: All values except n/G are basis points. Equal auction weights are used. Signed negative/positive/printed-zero counts are 142/92/6 in the primary sample, 118/79/6 in the no-row-flag subset, and 138/92/6 in the best-bid subset. These imply printed-zero shares 2.50%, 2.96% and 2.54%. Absolute outcomes preserve the same zeros. Descriptions were computed after the fits on the unchanged private input sets, not selected to improve estimates. The tables do not measure a treatment effect or procurement benefit.

S15 Cluster uncertainty, scale and numerical diagnostics

The primary uncertainty display uses quarter-cluster CR1 covariance with finite factor $G/(G-1) \times (n-1)/(n-\mathrm{k})$, and a t($G-1$) reference. The primary signed coefficient alone receives a restricted-null wild-cluster t test with 999 Rademacher sign vectors, SeedSequence([20261011, 2, 240]), and CR1 studentization. The null sets the focal coefficient to zero; no sign vector is redrawn following a failure. All 999 stored draws are valid. Four statistics meet or exceed the absolute observed statistic of 3.5632, giving the two-sided plus-one p-value (4+1)/(999+1) = 0.005. This is a test, not the source of a bootstrap confidence interval.

The quarter jackknife deletes each of 21 source quarters, recentres its coefficient estimates at their deletion mean, and uses variance $(G-1)/G$ times their summed squared deviations. This is mean-centred CV3J, distinct from the full-estimate-centred CV3. A t($G-1$) interval is displayed around the full-sample coefficient. The approaches follow the cluster-jackknife and wild-bootstrap framework discussed by MacKinnon et al. (2023); they remain approximate with few and potentially dependent time clusters. Agreement of these displays does not adjust the whole exploratory history or make the same observations independent replications.

Table S12 Distinct price scales and inference diagnostics

Specification	x-resid. IQR	Residual-IQR contrast [95% CI], bp	Scaled condition	Max. quarter share	Info. equiv. quarters
Primary: signed	0.1291	−10.16 [−16.11, −4.21]	14.76	0.0910	16.33
S1: absolute	0.1291	5.34 [−1.75, 12.43]	14.76	0.0910	16.33
S2: no row flags	0.1247	−10.16 [−17.26, −3.06]	17.46	0.1119	13.97
S3: best bid only	0.1270	−10.62 [−16.48, −4.77]	14.85	0.0921	16.15

Note: Residual-x contrasts differ from the observed-x contrasts in main Table 5. They are not alternatives selected by effect size. Information shares are proportional to within-model residualized x squared, summed by quarter; their reciprocal concentration is a design diagnostic, not a certified effective independent sample size. The primary observed-x contrast is −14.5059 bp; the residual-x contrast is −10.1622 bp. The primary CR1 SE is 22.0923; the CV3J SE is 27.7203 and its coefficient interval [−136.5430, −20.8960]. The single wild-cluster p-value is 0.005. Raw and scaled condition numbers address different numerical scales; no control was dropped to obtain significance.

S16 Source-subset and year-deletion limits

The four fixed extended specifications and their own identical-row baselines are reported in main Table 5. Only the primary model has the 999-draw wild-cluster test. The two source subsets overlap the primary data; their similar signs are not independent replications. The absolute-outcome coefficient is positive but imprecise, so neither a reduction in magnitude nor a zero effect has been established. No new natural-log price specification or further window search was added after viewing these results.

Table S13 Fixed leave-one-year-out primary price coefficients

Omitted year	Auctions retained	Quarters retained	κ (bp per unit x)
2020	194	17	−86.11
2021	195	17	−84.42
2022	205	18	−45.12
2023	204	18	−73.27
2024	206	18	−88.23
2025	196	17	−83.74

Note: These are point-estimate influence checks, not six significance tests. Observed-year effects are rebased after deletion; all economic controls are retained. Deleting 2022 reduces the coefficient magnitude from 78.72 to 45.12, a 42.7% comparison of two estimates, not an estimate of the contribution of that year. No claim is made that the deletion-specific intervals exclude zero or that slopes are invariant across years. The 21 quarter deletions and the four models remain available in the fixed RESULTS.json.

The price analysis used 1,034 estimation requests: eight baseline/extension fits, 999 restricted bootstrap fits, 21 quarter deletions and six year deletions. Four fixed extended models were also checked with a separate numerical implementation, which agreed on the focal coefficient, CR1 standard error and partial R^2. The recorded 26 artificial tests in each Python mode assess implementation behaviour rather than empirical validity. The 55 frozen source objects were unchanged. Public aggregate checks reproduce arithmetic from the outputs, but independent event-level refitting still requires the input rows.

S17 Report definitions, locators and availability

The targeted local source-definition audit covers 24 held quarterly reports: file identity and native text were checked, relevant Table 4 explanatory text was located, and the earliest and latest complete methods paragraphs were additionally read. That audit describes the target as a comparison of auction clearing prices with the secondary-market reference. It did not locate an explicit subtraction-and-percentage-denominator equation in the examined text. This does not prove that no formula occurs elsewhere or in an image. Accordingly, the article retains the official signed number and the phrase “lower official-reported signed deviation”, without reconstructing a numerator, dividing by an inferred reference price, or relabelling the association as a larger discount.

Table S14 Report-definition page locators and eligibility

Source quarter	Definition page	Reported auctions	Eligible auctions
2020Q1	13	12	12
2020Q2	14	10	10
2020Q3	14	13	13
2020Q4	15	11	11
2021Q1	13	9	9
2021Q2	14	11	11
2021Q3	15	13	13
2021Q4	16	12	12
2022Q1	13	11	11
2022Q2	14	10	10
2022Q3	15	14	14
2022Q4	15	11	0
2023Q1	13	12	12
2023Q2	13	11	11
2023Q3	14	13	13
2023Q4	15	11	0
2024Q1	13	10	10
2024Q2	14	12	0
2024Q3	14	13	13
2024Q4	15	11	11
2025Q1	14	12	11
2025Q2	15	10	10
2025Q3	15	13	13
2025Q4	16	10	10

Note: Page numbers identify the methods-text locations in the fixed source record. Online Resource 2 and the machine-readable SOURCE_DEFINITION_CHECK.json give exact source URLs and SHA-256 values. Three quarters contribute no eligible reference observations; one additional source-date conflict is excluded in 2025Q1. The reports and event-level panels are not redistributed with this resource. These locators do not authenticate native tick data, quote age or a complete UTC deadline history.

Source identities and reporting coverage are listed in Online Resource 2. Section S18 gives the public and conditional-access routes; neither the catalogue nor a file hash grants permission to redistribute third-party material.

S18 Data and code access

Data and code access proceeds in two stages. Public code, aggregate results, recorded bootstrap statistics and source metadata can be obtained through the fixed links below without requesting the event-level panels. Independent refitting requires the corresponding exact input rows. Requests about author-held matched tables or the 2026 quantity snapshot should be directed to Zhongfei Tian (15517837680@163.com); transfer of third-party files is subject to the relevant provider's permission. The source catalogue supports lawful retrieval from the original providers, but a

matching hash is required before treating a newly obtained file as the analysed snapshot. No unrestricted licence or guaranteed access to restricted material is implied.

Historical quantity inputs (fixed snapshot)

German price model, protocol and aggregate results

German price descriptions and source definitions

Historical quantity and extension reproduction map

Online Resource 2 (ESM_2.csv) lists the official report quarter, source URL, definition-page locator, eligibility count and file hash, with article and corresponding-author metadata. It contains no event-level price rows. The unit of the price input is numeric percentage points; the analysis converts this field to basis points without reconstructing native reference prices.

Additional references

German Emissions Trading Authority (DEHSt) (2020–2025) Quarterly reports on German auctions of emission allowances. Umweltbundesamt, Berlin. Report-specific editions, Table 4 page locators and source URLs are recorded in Online Resource 1, Section S17, and the accompanying source ledger.

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